

Scenario 4 - Casualties

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Casualties - Scenario 4

Traffic Accidents in the Netherlands
SAS Case Study

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This comprehensive analysis of traffic accidents in the Netherlands, facilitated by SAS® Visual Analytics, reveals key insights crucial for targeted road safety improvements. With a concentration of accidents around age 21, tailored safety initiatives for young adults are recommended. The case study identifies a significant correlation between accident types and hit-and-run incidents, emphasizing the need for enhanced enforcement and public awareness. High-risk areas, particularly straight roads, warrant focused safety measures. Pronounced provincial variations suggest the importance of localized campaigns, and recommendations include age-specific and day-specific interventions. These insights offer actionable strategies for policymakers and stakeholders to mitigate accident impact and enhance overall road safety in the Netherlands.

The Data

Casualty and Accident Data

Accident ending	Hour accident	Province	Province name	Road situation ▲	Time accident	Type accident	Type of casualty	Type of Casualty
Injury First-Aid	19	Drenthe	Drenthe		18:52	Unknown	Material	0.00
Injury Misc	12	Flevoland	Flevoland		12:00	Unknown	Material	0.00
Injury Misc	11	Limburg	Limburg		11:08	One-sided	Material	0.00
Injury Hospital	15	Gelderland	Gelderland		14:41	Unknown	Material	0.00
Injury Hospital	3	Gelderland	Gelderland		2:57	Unknown	Material	0.00
Injury Misc	19	Noord-Brabant	Noord-Brabant		19:00	One-sided	Material	0.00
Injury Misc	17	Zeeland	Zeeland		16:36	One-sided	Material	0.00
Injury Misc	2	Gelderland	Gelderland		2:20	Unknown	Material	0.00
Injury Misc	5	Noord-Holland	Noord-Holland		4:40	One-sided	Material	0.00
Injury Hospital	15	Friesland	Friesland		15:03	Unknown	Material	0.00
Injury Hospital	21	Gelderland	Gelderland		20:55	Unknown	Material	0.00
Injury Misc	19	Zuid-Holland	Zuid-Holland		19:29	One-sided	Material	0.00
Injury Hospital	14	Limburg	Limburg		13:48	Unknown	Material	0.00
Injury Hospital	24	Limburg	Limburg		23:38	Unknown	Material	0.00
Injury Hospital	13	Limburg	Limburg		12:39	One-sided	Material	0.00
Injury Hospital	10	Noord-Brabant	Noord-Brabant		9:50	Parked vehicle	Material	0.00
Injury Hospital	16	Noord-Brabant	Noord-Brabant		16:00	Unknown	Material	0.00
Injury Misc	1	Friesland	Friesland		0:44	Unknown	Material	0.00
Injury Hospital	15	Overijssel	Overijssel		14:59	Unknown	Material	0.00
Injury Hospital	15	Flevoland	Flevoland		14:40	Unknown	Material	0.00
Injury Hospital	15	Utrecht	Utrecht		14:58	Unknown	Material	0.00
Injury Hospital	21	Zeeland	Zeeland		20:45	Unknown	Material	0.00
Injury Hospital	11	Zuid-Holland	Zuid-Holland		10:33	Unknown	Material	0.00

Introduction

The case study revealed the causalities in traffic accidents in the Netherlands using SAS® Visual Analytics. It showed the traffic collision and safety standards adopted by traffic sector in Netherlands in order to make inform decisions to solve traffic collision and enhance safety standards among motorists and pedestrians.

The Dataset:

The dataset is made up of data collect for traffic accidents with a focus on the location of the accident, the time of the accident, the offenders and victims, causality information, and the type of accidents. The data contained more than 125,000 observations and 45 measures. However, Information based on who the causalities are contained 8 measure variables, 9 category variables and 1 geography variable.

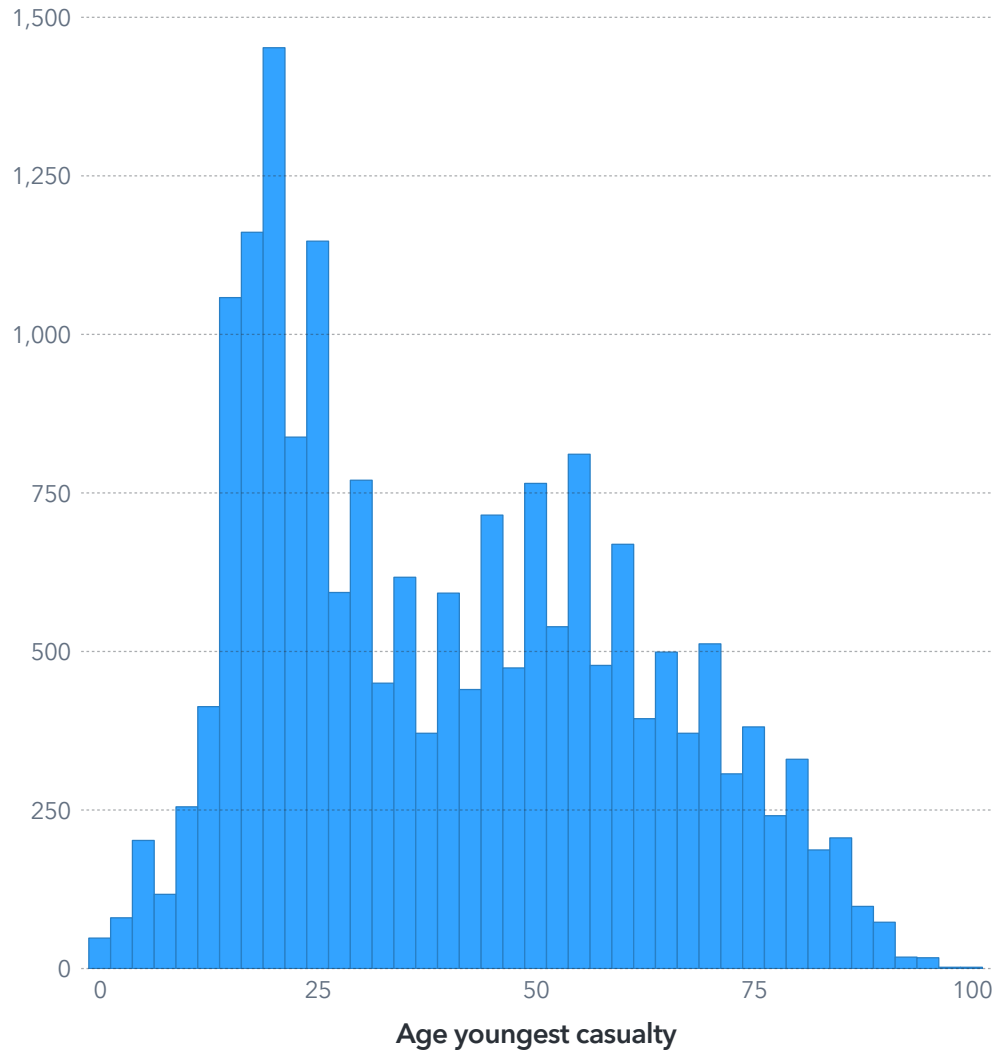
Data Exploration

The casualty data was first filtered to remove missing values for some category variable. After the filtering the dataset, some categorical variables such as the date accident, age youngest casualty, and type of accidents were categorized.

Histogram



Number accidents



A2.1

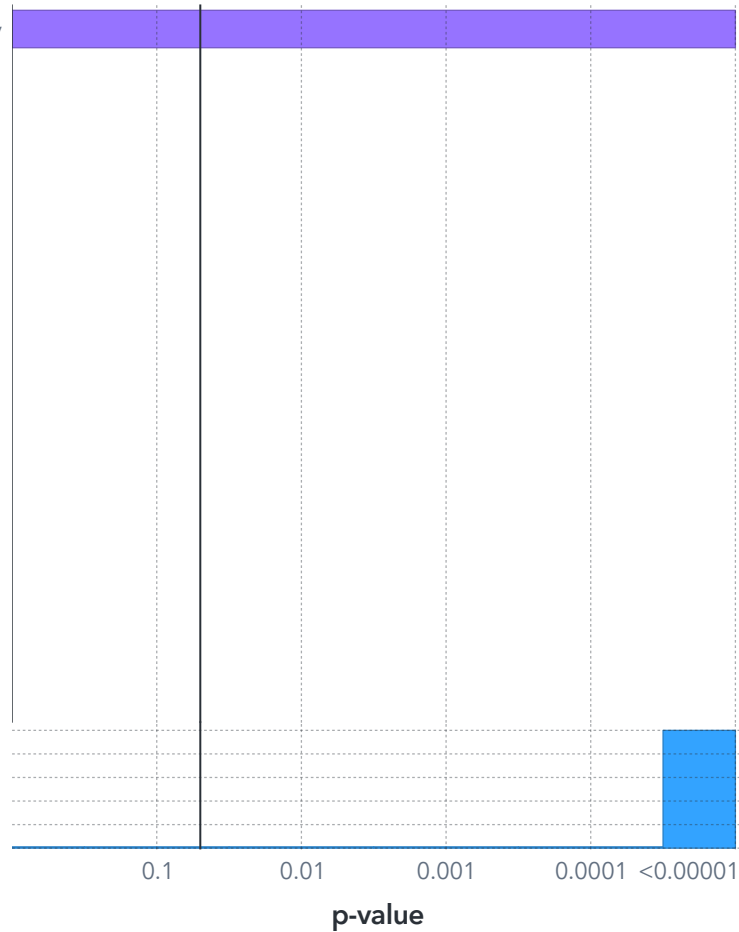
In the histogram of number of accidents by age of the casualty showed that more accidents occur at about age 21. The traffic accidents is prevalent among casualty within 19 years to 21 years.

Univariate Analysis

Linear Regression **Number casualties** R-Square **0.0352** Observations Used **18,693**

Fit Summary

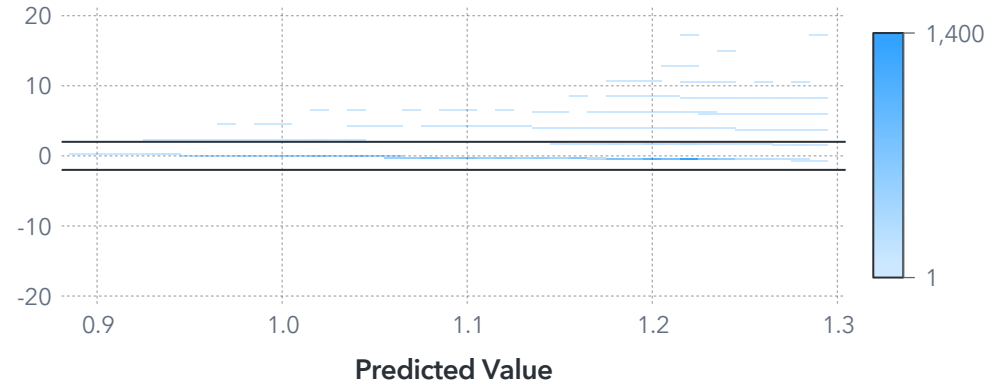
Age youngest casualty



A3.1

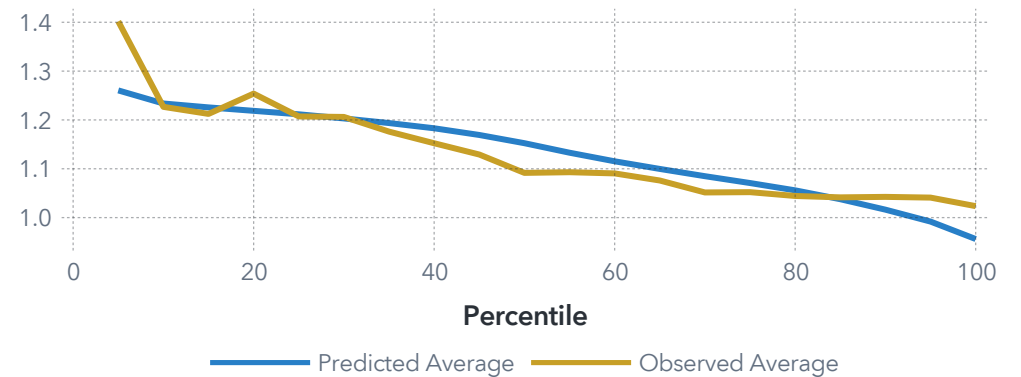
Residual Plot

Studentized Deleted Residual



Assessment

Number casualties



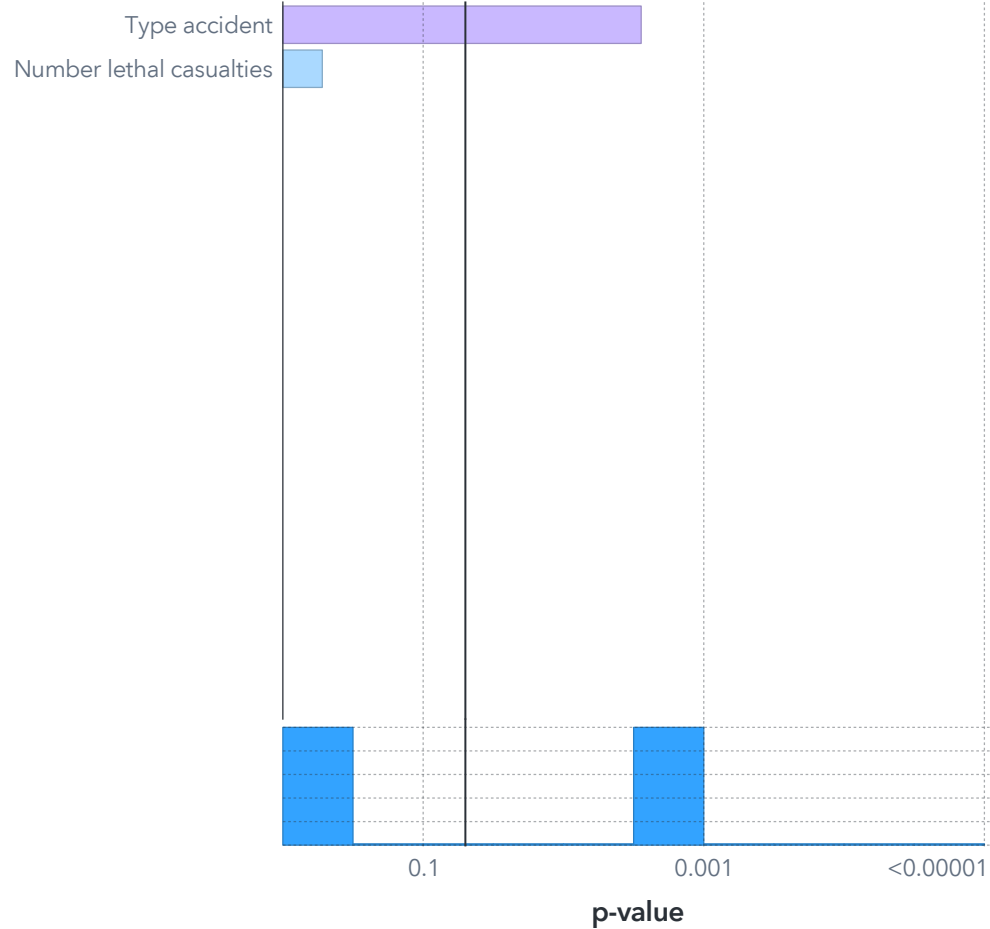
The univariate analysis showed the relationship between the ages of causality and the number of casualties in the accident. It was revealed that the ages of causality is statistically significant to the number of casualties.

Bivariate Analysis



Linear Regression **Number hit-and-run** Adjusted R-Square **0.0009** Observations Used **18,693**

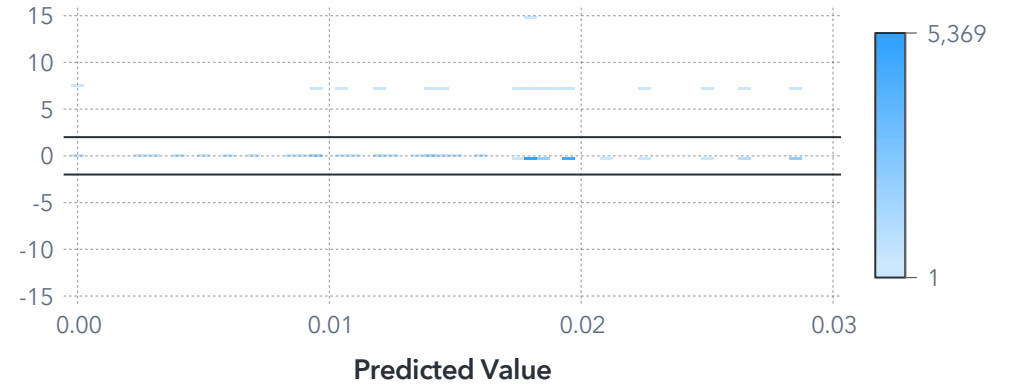
Fit Summary



A4.1

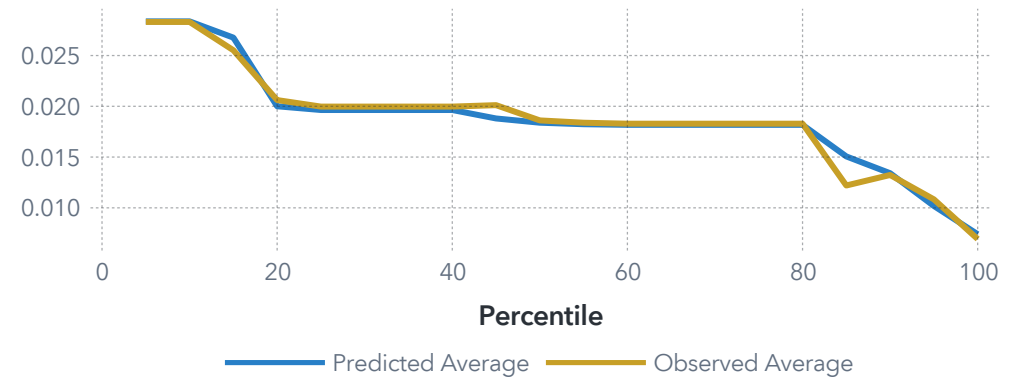
Residual Plot

Studentized Deleted Residual



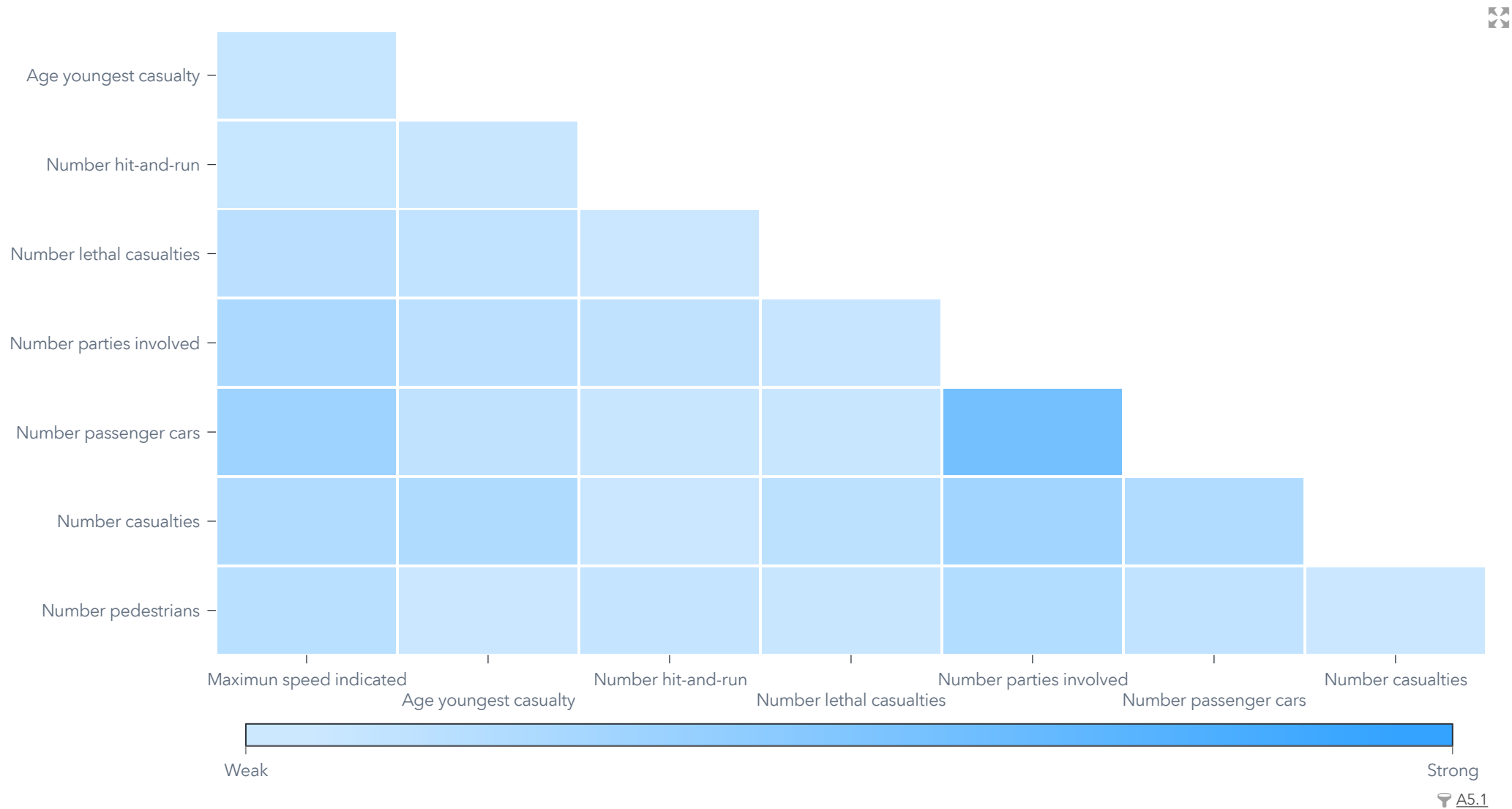
Assessment

Number hit-and-run



The bivariate analysis showed that type of accident predict the number of hit-and-run while the number lethal casualties is not significant to the number of hit-and-run.

Correlation Matrix

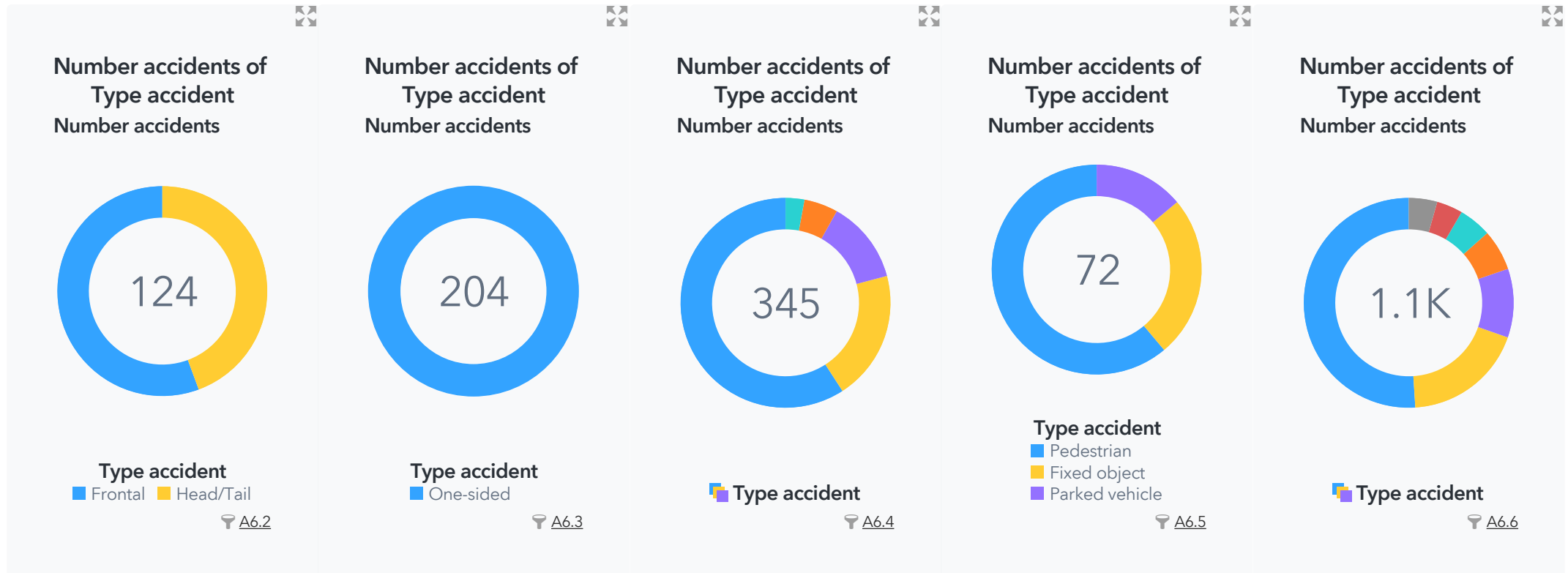


The correlational matrix showed the relationship between the accidents and casualties variables. There is a moderate relationship number passenger car and the number of parties involved is moderate, while maximum speed indicated showed a weak relationship with number of lethal casualties, and a weak relationship between the number of parties involved and the number of casualties.

Type of Accidents by Province

Bend	Crossing 3-way	Crossing 4-way	Roundabout	Straight road
------	----------------	----------------	------------	---------------

A6.1



Drenthe	Flevoland	Friesland	Gelderland	Groningen	Limburg	Noord-Brabant	Noord-Holland	Overijssel	Utrecht	Zeeland	Zuid-Holla
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A6.7

The pie chart showed the relationship number of accidents by the type of accident based on the province where the accidents took place. Drenthe province recorded 80 accidents involving 37 head/tail, 4 animal accident, and 39 frontal accident. On bend road in Friesland province, there was 17 fixed object accident, 30 one-sided accident, 10 frontal accident, and 5 side accidents. On roundabout road in Groningen province, there were 3 one-sided accidents within the province, 1 pedestrian accidents. The province of Noord-Brabant recorded 175 head/tail accidents on a straight road, and 68 frontal accident, 253 one-sided accident.

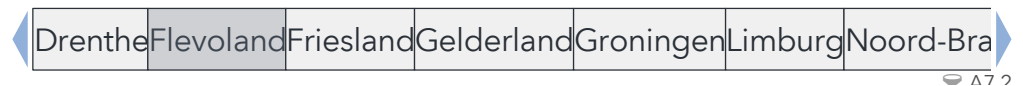
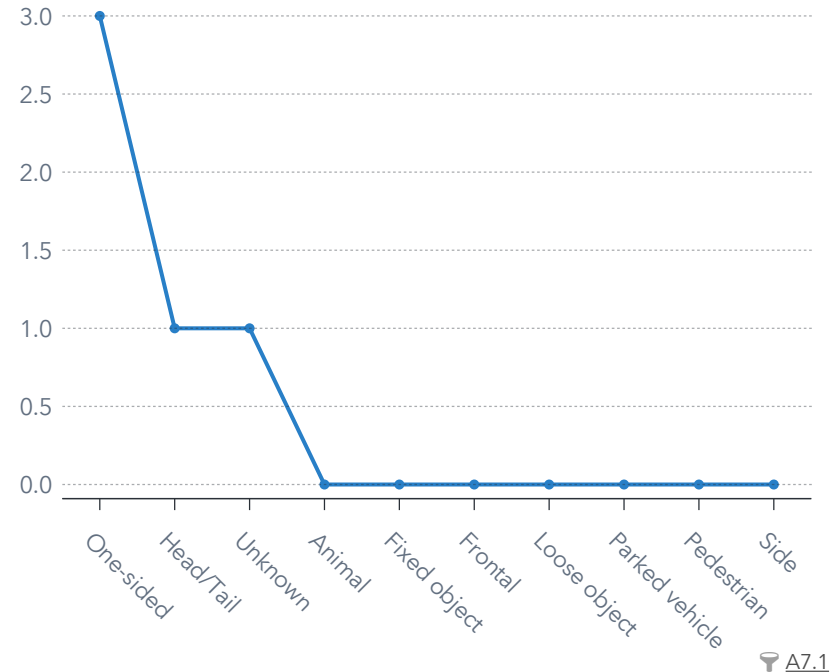
Graphical Representation Comparing Different Accident/Casualty Variables

In Drenthe province, the number hit-and-run by type of accident recorded no accidents in animal, head/tail, loose object, parked vehicle, pedestrian, side and unknown accident. In the Gelderland province, the highest type of hit-and-run accident is the side accident while animal, loose object and parked vehicle recorded the lowest number of hit-and-run accidents.

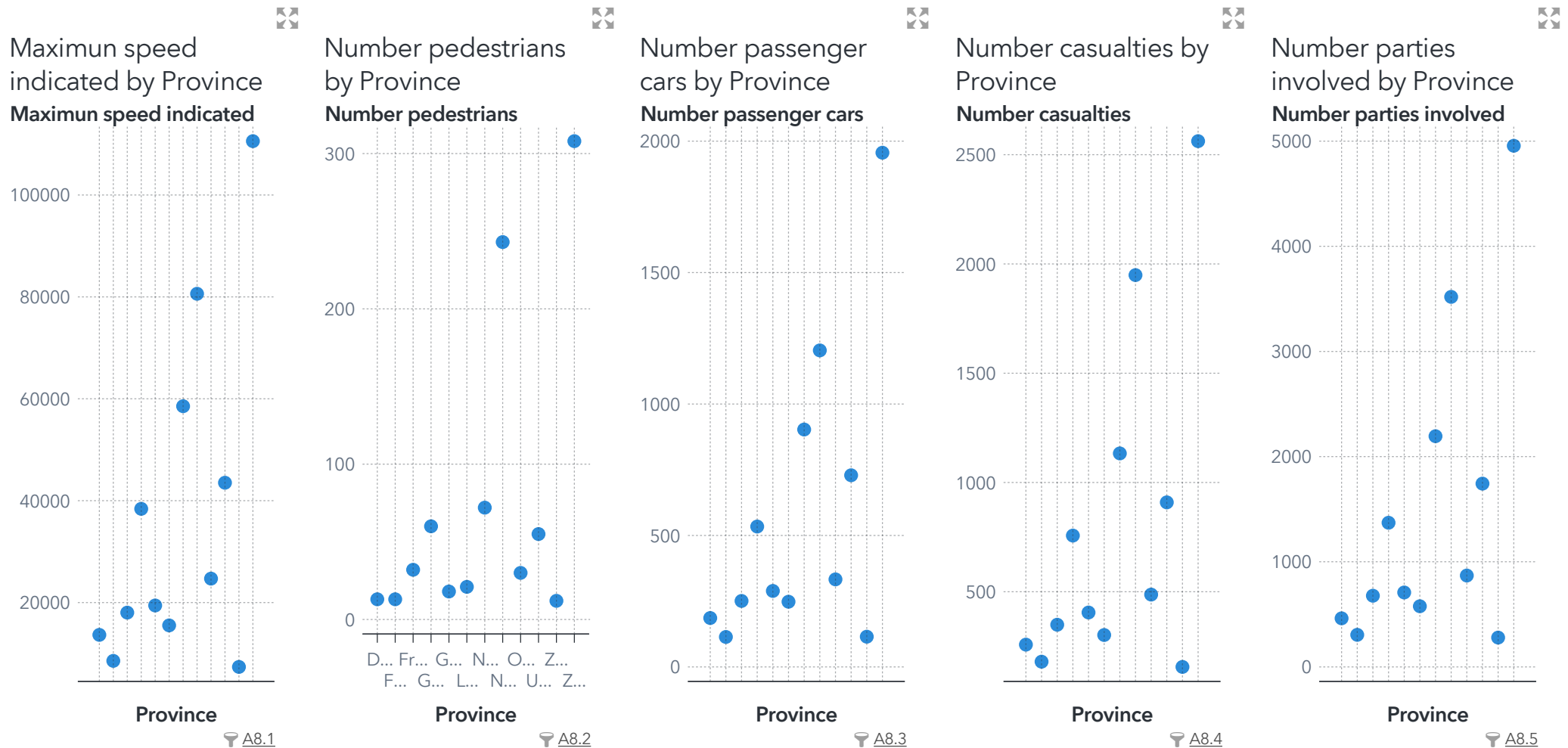
A common trend among the provinces revealed that one-sided accident has the highest number of casualties followed by side accident. There were more number of material casualty than injured casualty in Flevoland province, while lethal type of casualty was low in Limburg province.

The relationship between the number accidents, number lethal casualties by the day of accident showed that the highest number of accident, lethal casualties was recorded on Friday and Saturday in Drenthe province, while in Flevoland province, the highest number of accident, lethal casualties was recorded on Thursday while the lowest number of accident, lethal casualties was recorded on Monday, Friday and Saturday.

Number hit-and-run by Type accident



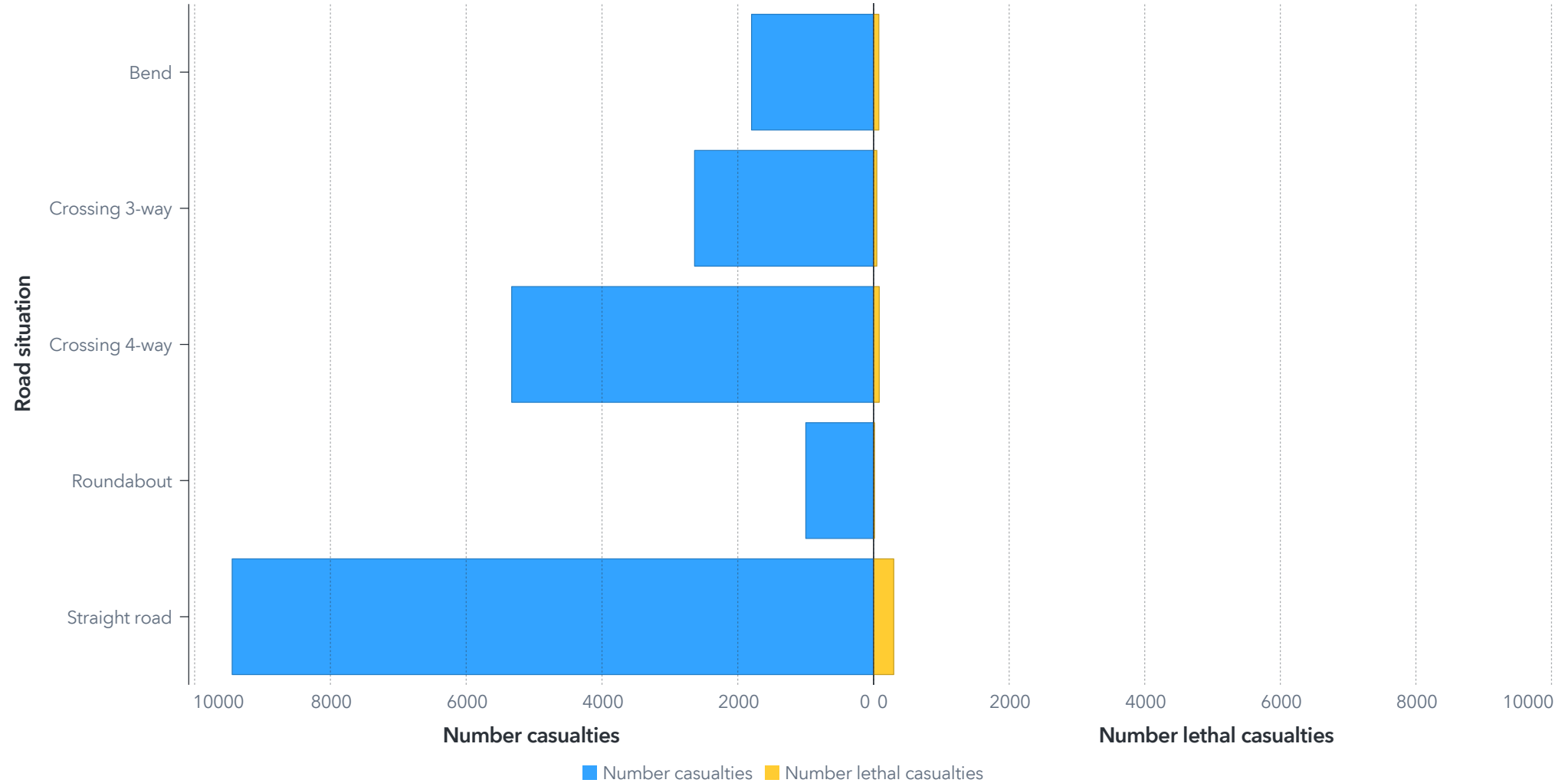
Relationship between Accident/Casualty Variables based on Provinces



Noord-Brabant recorded the highest maximum speed on a bend, Noord-Holland recorded the highest number of pedestrians on a bend, Zuid-Holland recorded the highest of casualties on a bend road. Likewise, Zuid-Holland recorded the highest maximum speed and the highest number of pedestrians, casualties and the number of parties involved in accidents on a crossing 4-way road. In a roundabout, the province of Noord-Holland recorded the highest number of pedestrians, casualties, parties involved, and the maximum speed by motorists. In the same vein, the province of Noord-Holland recorded the highest number of pedestrians, casualties, parties involved, passenger cars, and the maximum speed by motorists.

Relationship between Number of casualties by Road sSituation

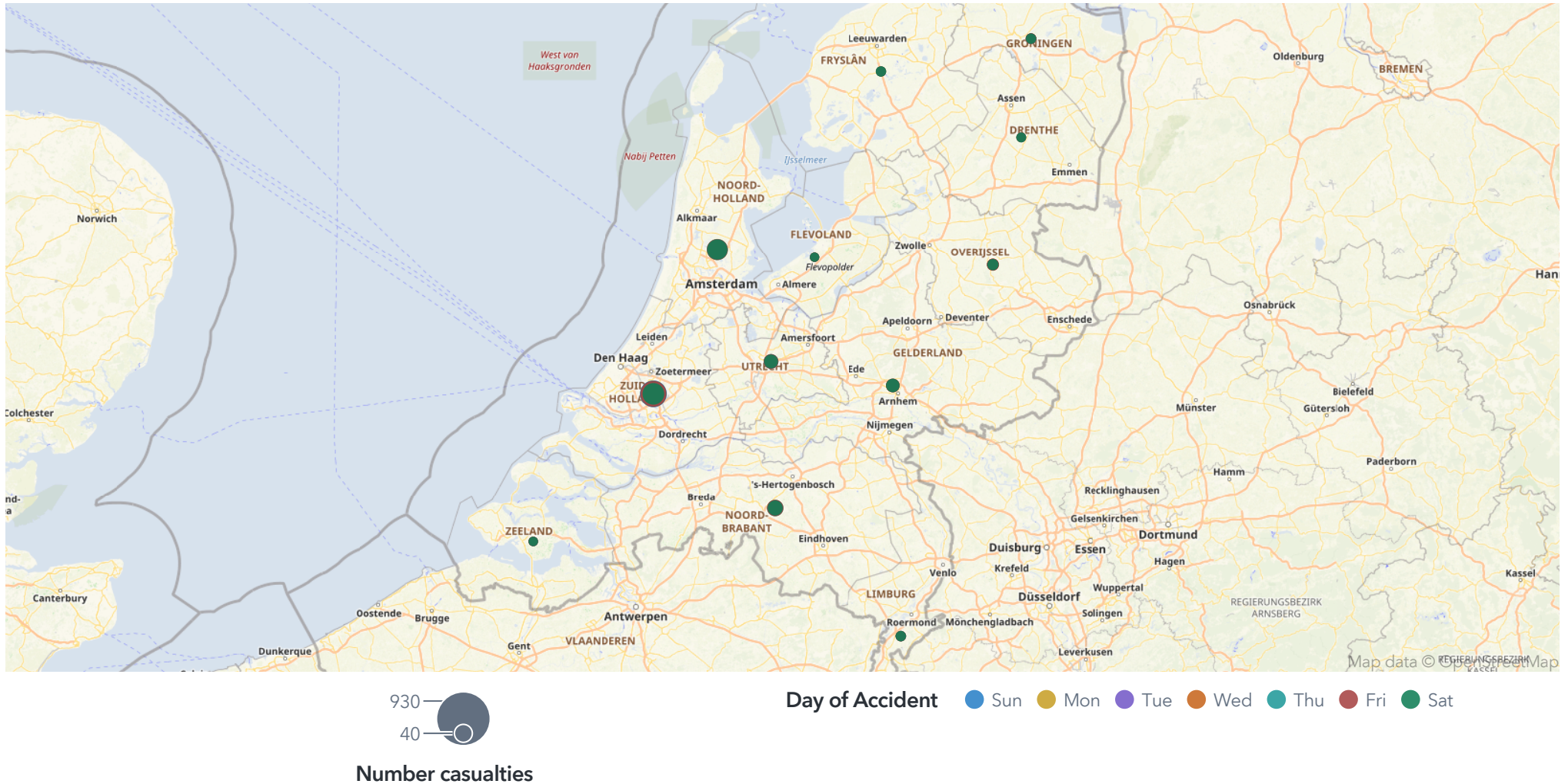
Number casualties, Number lethal casualties by Road situation



Among all the road situations, straight road recorded the highest number of casualties, and lethal (fatal) casualties for accident victims while roundabout recorded the least number of casualties and lethal casualties for accident victims.

Geo Cordinate Showing Week of Accidents by Number of Casualties based on Province

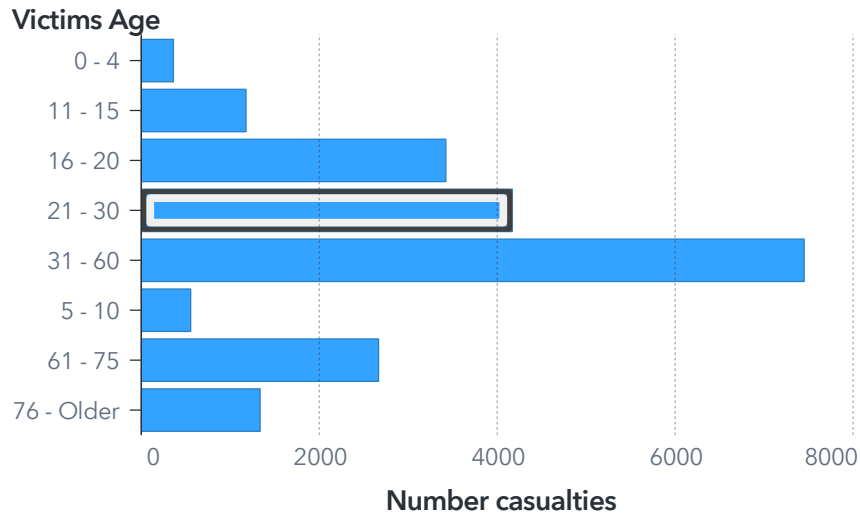
Day of Accident by Province sized by Number casualties



Zuid-Holland recorded the highest number of accident casualties among all the provinces in a week while Flevoland recorded the least number of accident casualties among all the provinces in a week.

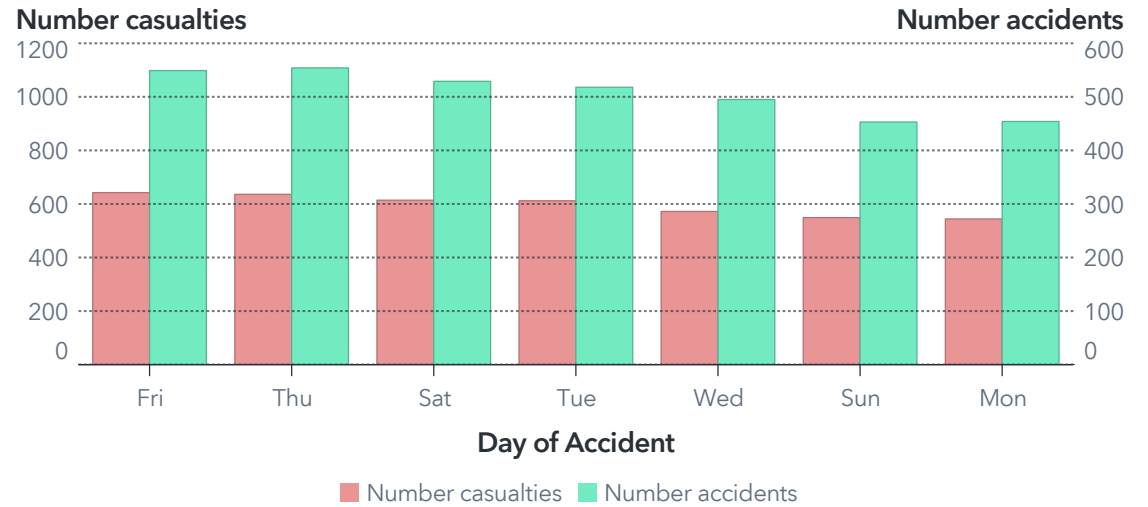
Casualties

Number casualties by Victims Age



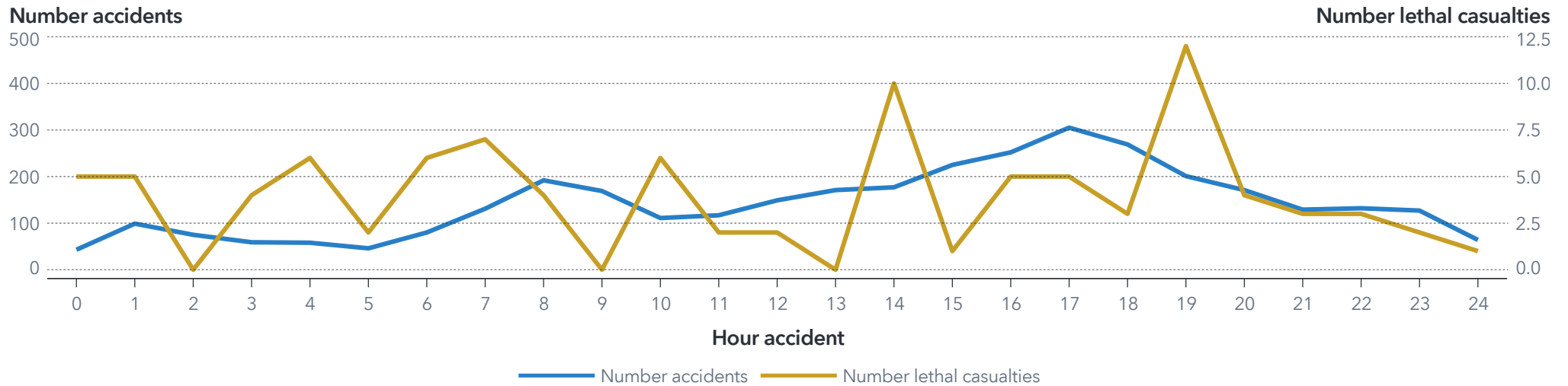
A11.1

Number casualties, Number accidents by Day of Accident



A11.2

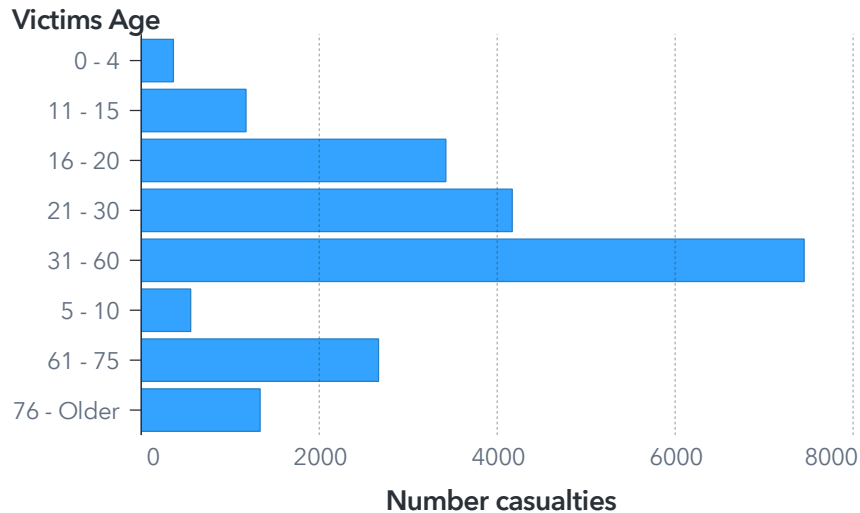
Number accidents, Number lethal casualties by Hour accident



A11.3

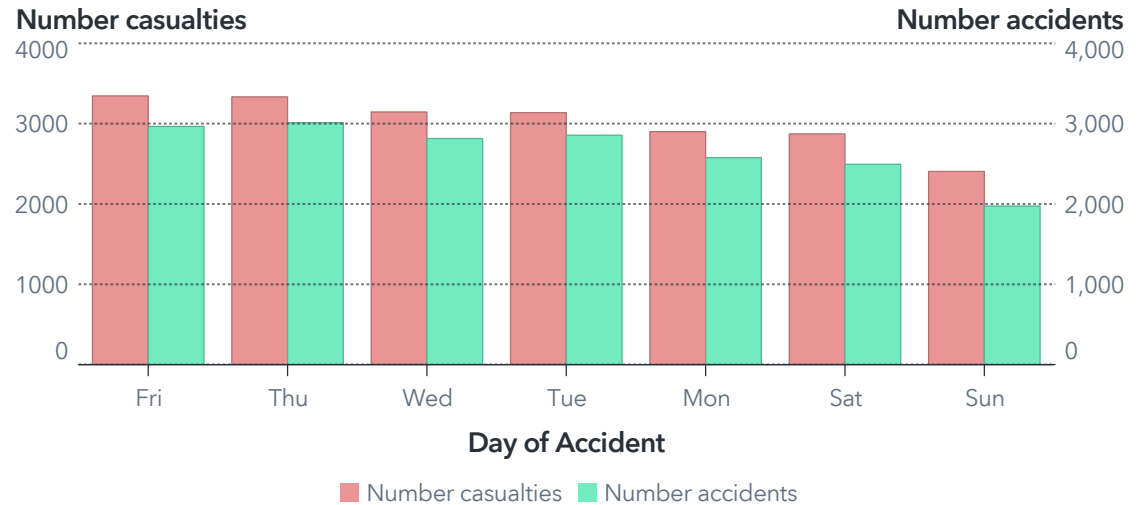
Visual Solution with Explanation

Number casualties by Victims Age



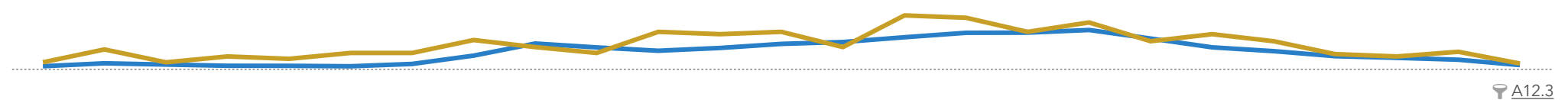
A12.1

Number casualties, Number accidents by Day of Accident



A12.2

Number accidents, Number lethal casualties by Hour accident



A12.3

The comprehensive analysis of the accident scenario revealed the number of casualties in the traffic accidents in the Netherlands. It also revealed the victims' age, the day of the accidents and the time the accident took place. The relationship between number casualties by victims' age showed that motorists/pedestrians within age of 31-60 are the highest casualties in the traffic accident.

The highest number of casualties and accidents were recorded on Fridays while the least number of casualties and accidents were recorded on Sunday irrespective of the victims' age.

The highest number of accidents, lethal casualties in traffic accidents were recorded at 17:00 Hours for victims within the age range of 5 -10 years while victims with the age of 31 -60 years recorded the highest number of accidents, lethal casualties in traffic accidents at 20:00 and 17:00 Hours respectively. For older accident victims within the age of 76 and above, the highest number of accidents, lethal casualties in traffic accidents were recorded at 21:00 hours respectively.

Conclusion

Using SAS® Visual Analytics to analyze traffic incidents in the Netherlands produced important insights into a number of different areas of road safety. Interestingly, the data showed that there was a statistically significant association between the age of the casualties and the total number of casualties, and that the accidents were concentrated around the age of 21. While there was no discernible relationship between the number of fatalities and hit-and-run episodes, the case study did provide insight into the predictive ability of accident types in foretelling hit-and-run incidents. An example of a moderate relationship between the number of passenger automobiles and parties engaged in accidents was highlighted by the correlational matrix.

Comprehensive provincial analysis also showed significant differences in accident categories, fatalities, and time trends. For example, Noord-Brabant showed some patterns, such as a high frequency of "head/tail" incidents on straight roads. Regardless of the age of the victims, Fridays were regularly recorded as having the largest number of accidents and casualties, according to the temporal study. Together, these data present a comprehensive picture of the dynamics of traffic accidents in the Netherlands, which is helpful for stakeholders and policymakers who want to improve road safety regulations and reduce particular hazards related to certain variables and geographical areas.

Recommendations

The following are recommendations:

- 1. Specific Safety Measures for Young Adults:** There is an obvious need for focused safety programs addressed at young adults, especially those between the ages of 19 and 21, given the concentration of road accidents around the age of 21. It would be possible to address the particular hazards and behaviors causing accidents in this age range by implementing educational programs, awareness campaigns, and strengthened law enforcement measures.
- 2. Strengthened Implementation to Combat Hit-and-Run Occurrences:** The data emphasizes how important different accident types are in forecasting hit-and-run instances. Law enforcement organizations should think about introducing tighter sanctions, increased surveillance, and public awareness campaigns that particularly address the repercussions of leaving the scene in order to reduce these kinds of incidents. To discourage hit-and-run activity, law enforcement and local communities can work together to great effect.
- 3. Focus on High-Risk Road Situations:** Straight highways typically record the largest number of fatal occurrences and casualties, as the analysis has shown. This emphasizes how crucial it is to put specific safety measures in place on these kinds of roads, like better signage, speed limit enforcement, and infrastructure upgrades, in order to increase overall safety and lessen the severity of incidents.
- 4. Day-Specific Traffic Safety Measures:** Fridays consistently see the highest number of accidents and fatalities, so there's a chance to use day-specific traffic safety protocols. This might entail stepped-up law enforcement presence on Fridays, public awareness campaigns, and focused initiatives meant to curb risky behaviors. Adapting interventions to particular days of the week can help develop accident prevention plans that are more successful.
- 5. Provide Safety Initiatives Exclusive to a Province:** The provincial variations in accident types and patterns suggest the need for localized safety campaigns and interventions. Tailoring safety measures to address specific challenges in each province, as highlighted by the analysis, can lead to more effective outcomes. Collaboration between local authorities, transportation agencies, and community stakeholders is crucial for the successful implementation of these provincial-specific safety initiatives.

Appendix

A1.1 Casualty and Accident Data

Warnings: Not all of the data appears because the client query row limit exceeds the server threshold.

Filters: Age youngest casualty NOT MISSING

A2.1 Histogram - Age youngest casualty 1

Filters: Age youngest casualty NOT MISSING

A3.1 Fit Summary

Filters: Age youngest casualty NOT MISSING

A4.1 Fit Summary

Filters: Age youngest casualty NOT MISSING

A5.1 Correlation - Maximun speed indicated 1

Filters: Age youngest casualty NOT MISSING

A6.1 Button bar - Road situation 1

Filters: Age youngest casualty NOT MISSING
Province NOT MISSING
Road situation NOT MISSING

A6.2 Number accidents of Type accident

Filters: Age youngest casualty NOT MISSING
NOT(Type accident = 'Fixed object') AND NOT((Type accident = 'Loose object') OR (Type accident = 'Parked vehicle') OR (Type accident = 'Pedestrian')) AND NOT(Type accident = 'Side') AND NOT(Type accident = 'One-sided') AND NOT(Type accident = 'Unknown')
Road situation = 'Crossing 4-way'
Province = 'Noord-Holland'

A6.3 Number accidents of Type accident

Filters: Age youngest casualty NOT MISSING
Type accident = 'One-sided'
Road situation = 'Crossing 4-way'
Province = 'Noord-Holland'

A6.4 Number accidents of Type accident

Filters: Age youngest casualty NOT MISSING
NOT((Type accident = 'Head/Tail') OR (Type accident = 'Side') OR (Type accident = 'Unknown') OR (Type accident = 'Loose object'))
Road situation = 'Crossing 4-way'
Province = 'Noord-Holland'

A6.5 Number accidents of Type accident

Filters: Age youngest casualty NOT MISSING
NOT(Type accident = 'Side') AND NOT(Type accident = 'One-sided') AND NOT(Type accident = 'Frontal') AND NOT(Type accident = 'Unknown') AND NOT(Type accident = 'Head/Tail') AND NOT(Type accident = 'Loose object')
Road situation = 'Crossing 4-way'
Province = 'Noord-Holland'

A6.6 Number accidents of Type accident

Filters: Age youngest casualty NOT MISSING
Road situation = 'Crossing 4-way'
Province = 'Noord-Holland'

A6.7 Button bar - Province 1

Filters: Age youngest casualty NOT MISSING

A7.1 Number hit-and-run by Type accident

Filters: Age youngest casualty NOT MISSING
Province = 'Flevoland'

A7.2 Button bar - Province 2

Filters: Age youngest casualty NOT MISSING

A8.1 Maximun speed indicated by Province

Filters: Age youngest casualty NOT MISSING
Road situation = 'Straight road'

A8.2 Number pedestrians by Province

Filters: Age youngest casualty NOT MISSING
Road situation = 'Straight road'

A8.3 Number passenger cars by Province

Filters: Age youngest casualty NOT MISSING
Road situation = 'Straight road'

A8.4 Number casualties by Province

Filters: Age youngest casualty NOT MISSING
Road situation = 'Straight road'

A8.5 Number parties involved by Province

Filters: Age youngest casualty NOT MISSING
Road situation = 'Straight road'

A9.1 Number casualties, Number lethal casualties by Road situation

Filters: Age youngest casualty NOT MISSING
NOT(Road situation = ' ')

A10.1 Day of Accident by Province sized by Number casualties

Filters: Age youngest casualty NOT MISSING

A11.1 Number casualties by Victims Age

Filters: Age youngest casualty NOT MISSING
Victims Age IN { ALL }

A11.2 Number casualties, Number accidents by Day of Accident

Filters: Age youngest casualty NOT MISSING
Victims Age = '21 - 30'

A11.3 Number accidents, Number lethal casualties by Hour accident

Filters: Age youngest casualty NOT MISSING
Victims Age = '21 - 30'

A12.1 Number casualties by Victims Age

Filters: Age youngest casualty NOT MISSING
Victims Age IN { ALL }

A12.2 Number casualties, Number accidents by Day of Accident

Filters: Age youngest casualty NOT MISSING

A12.3 Number accidents, Number lethal casualties by Hour accident

Filters: Age youngest casualty NOT MISSING

Age youngest casualty (lower)	Age youngest casualty (upper)	Number accidents
-1	1	48
1	4	80
4	6	202
6	9	117
9	11	255
11	14	413
14	16	1,058
16	19	1,161
19	21	1,452
21	24	838
24	26	1,147
26	29	593
29	31	770
31	34	450
34	36	617
36	39	371
39	41	592
41	44	440
44	46	715
46	49	474
49	51	765
51	54	539
54	56	811
56	59	478
59	61	669
61	64	394
64	66	499
66	69	371
69	71	512
71	74	307
74	76	381
76	79	241
79	81	330
81	84	187
84	86	206
86	89	98
89	91	73
91	94	18
94	96	17
96	99	2
99	101	2

Description	Value	✕✕
Number of Model Effects	2	
Number of Classification Effects	0	
Number of Columns in X	2	
Rank of Cross-product Matrix	2	
Number of Observations Read	18,693	
Number of Observations Used	18,693	

Linear regression - Number casualties 1 Supplement 2

Source	Deg Freedom	Sum of Squares	Mean Square	F Value	Pr > F	R-Square
Model	1	139.5798	139.5798	682.2144	<0.00001	0.035214
Error	18691	3824.143	0.204598	.	.	.
Corrected Total	18692	3963.723	.	.	.	.

Statistic	Value	✕✕
Mean Square Error	0.204598	
Root MSE	0.452325	
F Value for Model	682.2144	
Pr > F	<0.00001	
R-Square	0.035214	
Adjusted R-square	0.035163	
AIC	-10963.3	
AICC	-10963.3	
SBC	-29642.7	

Linear regression - Number casualties 1 Supplement 4

Parameter	Estimate	Standard Error	t Value	Pr > t
Intercept	1.292804	0.007033	183.8107	<0.00001
Age youngest casualty	-0.00402	0.000154	-26.1192	<0.00001

Linear regression - Number casualties 1 Supplement 5

Effect	DF	DenDF	F Value	Pr > F
Age youngest casualty	1	18,691	682.2144	<0.00001

Percentile	Observations	Predicted Average	Observed Average
5	935	1	1
10	935	1	1
15	935	1	1
20	935	1	1
25	935	1	1
30	935	1	1
35	935	1	1
40	935	1	1
45	935	1	1
50	935	1	1
55	935	1	1
60	935	1	1
65	935	1	1
70	935	1	1
75	935	1	1
80	935	1	1
85	935	1	1
90	935	1	1
95	935	1	1
100	928	1	1

Linear regression - Number casualties 1 Supplement 7

ASE	Observed Average	SSE	Observations Used	Unused
0.2046	1.4019	3,824.1430	18,693	0

Description	Value	✕✕
Number of Model Effects	3	
Number of Classification Effects	1	
Number of Columns in X	12	
Rank of Cross-product Matrix	11	
Number of Observations Read	18,693	
Number of Observations Used	18,693	

Linear regression - Number hit-and-run 1 Supplement 2

Source	Deg Freedom	Sum of Squares	Mean Square	F Value	Pr > F	R-Square
Model	10	0.479943	0.047994	2.599178	0.00377	0.001389
Error	18682	344.9668	0.018465	.	.	.
Corrected Total	18692	345.4467	.	.	.	.

Statistic	Value	✕✕
Mean Square Error	0.018465	
Root MSE	0.135887	
F Value for Model	2.599178	
Pr > F	0.00377	
R-Square	0.001389	
Adjusted R-square	0.000855	
AIC	-55914	
AICC	-55914	
SBC	-74522.8	

Parameter	Estimate	Standard Error	t Value	Pr > t
Intercept	0.028365	0.002953	9.605705	<0.00001
Type accident Animal	-0.01106	0.018085	-0.61162	0.54080
Type accident Fixed object	-0.01882	0.004805	-3.91708	0.00009
Type accident Frontal	-0.01439	0.004869	-2.95506	0.00313
Type accident Head/Tail	-0.00998	0.00432	-2.31042	0.02088
Type accident Loose object	-0.02458	0.008766	-2.80407	0.00505
Type accident One-sided	-0.00873	0.003521	-2.47977	0.01316
Type accident Parked vehicle	-0.01626	0.006767	-2.40274	0.01628
Type accident Pedestrian	-0.00208	0.005731	-0.36223	0.71718
Type accident Side	-0.01019	0.003467	-2.93906	0.00330
Type accident Unknown	0	.	.	.
Number lethal casualties	-0.00361	0.00566	-0.63798	0.52349

Linear regression - Number hit-and-run 1 Supplement 5

Effect	DF	DenDF	F Value	Pr > F
Type accident	9	18,682	2.797465	0.00279
Number lethal casualties	1	18,682	0.407022	0.52349

Percentile	Observations	Predicted Average	Observed Average
5	935	0.0283649118	0.0283109405
10	935	0.0283649118	0.0283109405
15	935	0.0267639777	0.0255197972
20	935	0.0200018373	0.0206129809
25	935	0.0196344895	0.0199714693
30	935	0.0196344895	0.0199714693
35	935	0.0196344895	0.0199714693
40	935	0.0196344895	0.0199714693
45	935	0.0188015725	0.0201137523
50	935	0.0183833491	0.0186097427
55	935	0.0182354702	0.0183810678
60	935	0.0181754292	0.0182882224
65	935	0.0181754292	0.0182882224
70	935	0.0181754292	0.0182882224
75	935	0.0181754292	0.0182882224
80	935	0.0181754292	0.0182882224
85	935	0.0150649807	0.0121991279
90	935	0.0134093297	0.0132358548
95	935	0.0101816439	0.0108343693
100	928	0.0073998622	0.0069278539

Linear regression - Number hit-and-run 1 Supplement 7

ASE	Observed Average	SSE	Observations Used	Unused
0.0185	0.0283	344.9668	18,693	0

X Axis	Y Axis	Correlation 
Maximun speed indicated	Number pedestrians	-0.1150
Maximun speed indicated	Number casualties	0.1640
Maximun speed indicated	Number passenger cars	0.2963
Maximun speed indicated	Number parties involved	0.2046
Maximun speed indicated	Number lethal casualties	0.1054
Maximun speed indicated	Number hit-and-run	-0.0279
Maximun speed indicated	Age youngest casualty	-0.0342
Age youngest casualty	Number pedestrians	0.0014
Age youngest casualty	Number casualties	-0.1877
Age youngest casualty	Number passenger cars	-0.0772
Age youngest casualty	Number parties involved	-0.1043
Age youngest casualty	Number lethal casualties	0.0658
Age youngest casualty	Number hit-and-run	-0.0150
Number hit-and-run	Number pedestrians	0.0486
Number hit-and-run	Number casualties	-0.0023
Number hit-and-run	Number passenger cars	-0.0145
Number hit-and-run	Number parties involved	0.0743
Number hit-and-run	Number lethal casualties	-0.0066
Number lethal casualties	Number pedestrians	0.0160
Number lethal casualties	Number casualties	0.0997
Number lethal casualties	Number passenger cars	0.0190
Number lethal casualties	Number parties involved	0.0324
Number parties involved	Number pedestrians	0.1593
Number parties involved	Number casualties	0.2722
Number parties involved	Number passenger cars	0.5737
Number passenger cars	Number pedestrians	-0.0649
Number passenger cars	Number casualties	0.1738
Number casualties	Number pedestrians	0.0049

Pie - Type accident 3 Supplement 1

Type accident▲	Number accidents✪✪
Frontal	69
Head/Tail	55

Pie - Type accident 2 Supplement 1

Type accident▲	Number accidents★▼
One-sided	204

Pie - Type accident 4 Supplement 1

Type accident ▲	Number accidents ▼
One-sided	204
Frontal	69
Pedestrian	44
Fixed object	18
Parked vehicle	10

Pie - Type accident 1 Supplement 1

Type accident ▲	Number accidents ▼
Pedestrian	44
Fixed object	18
Parked vehicle	10

Pie - Type accident 5 Supplement 1

Type accident▲	Number accidents▼▼
Side	558
One-sided	204
Unknown	116
Frontal	69
Head/Tail	55
Pedestrian	44
Loose object	20
Fixed object	18
Parked vehicle	10

Type accident▲	Number hit-and-run▼
One-sided	3
Head/Tail	1
Unknown	1
Animal	0
Fixed object	0
Frontal	0
Loose object	0
Parked vehicle	0
Pedestrian	0
Side	0

Province	Maximun speed indicated
Drenthe	13660
Flevoland	8560
Friesland	18030
Gelderland	38400
Groningen	19430
Limburg	15520
Noord-Brabant	58540
Noord-Holland	80610
Overijssel	24700
Utrecht	43520
Zeeland	7370
Zuid-Holland	110570

Province ▲	Number pedestrians ✖✖
Drenthe	13
Flevoland	13
Friesland	32
Gelderland	60
Groningen	18
Limburg	21
Noord-Brabant	72
Noord-Holland	243
Overijssel	30
Utrecht	55
Zeeland	12
Zuid-Holland	308

Province	Number passenger cars
Drenthe	186
Flevoland	114
Friesland	251
Gelderland	534
Groningen	289
Limburg	248
Noord-Brabant	903
Noord-Holland	1204
Overijssel	333
Utrecht	729
Zeeland	115
Zuid-Holland	1956

Province	Number casualties
Drenthe	258
Flevoland	180
Friesland	349
Gelderland	757
Groningen	405
Limburg	302
Noord-Brabant	1133
Noord-Holland	1949
Overijssel	487
Utrecht	909
Zeeland	156
Zuid-Holland	2562

Province ▲	Number parties involved ✖✖
Drenthe	462
Flevoland	305
Friesland	677
Gelderland	1371
Groningen	708
Limburg	577
Noord-Brabant	2194
Noord-Holland	3519
Overijssel	869
Utrecht	1743
Zeeland	279
Zuid-Holland	4956

Butterfly - Road situation 1 Supplement 1

Road situation ▲	Number casualties	Number lethal casualties ✖✖
Bend	1797	78
Crossing 3-way	2636	48
Crossing 4-way	5330	84
Roundabout	998	7
Straight road	9447	298

Province	Number casualties	Day of Accident	⌵⌵
Groningen	103	Sun	
Limburg	72	Sun	
Overijssel	140	Sun	
Drenthe	63	Sun	
Friesland	96	Sun	
Zuid-Holland	582	Sun	
Noord-Holland	490	Sun	
Utrecht	208	Sun	
Gelderland	194	Sun	
Noord-Brabant	347	Sun	
Flevoland	40	Sun	
Zeeland	70	Sun	
Zuid-Holland	734	Mon	
Groningen	129	Mon	
Drenthe	67	Mon	
Flevoland	56	Mon	
Utrecht	263	Mon	
Overijssel	174	Mon	
Noord-Brabant	397	Mon	
Zeeland	56	Mon	
Friesland	92	Mon	
Gelderland	279	Mon	
Limburg	103	Mon	
Noord-Holland	550	Mon	
Groningen	132	Tue	
Overijssel	178	Tue	
Noord-Holland	615	Tue	
Gelderland	220	Tue	
Drenthe	73	Tue	
Noord-Brabant	356	Tue	
Limburg	110	Tue	
Zeeland	70	Tue	
Friesland	85	Tue	
Flevoland	50	Tue	
Utrecht	317	Tue	
Zuid-Holland	930	Tue	
Overijssel	186	Wed	
Noord-Holland	657	Wed	
Gelderland	264	Wed	
Zeeland	69	Wed	
Noord-Brabant	353	Wed	
Flevoland	72	Wed	
Zuid-Holland	845	Wed	
Limburg	107	Wed	
Friesland	114	Wed	
Drenthe	73	Wed	
Utrecht	273	Wed	
Groningen	132	Wed	
Overijssel	170	Thu	
Noord-Holland	646	Thu	
Gelderland	273	Thu	
Flevoland	74	Thu	
Noord-Brabant	420	Thu	
Drenthe	65	Thu	
Zuid-Holland	924	Thu	
Limburg	119	Thu	
Friesland	101	Thu	
Zeeland	78	Thu	
Utrecht	334	Thu	
Groningen	129	Thu	
Overijssel	209	Fri	
Noord-Holland	624	Fri	
Gelderland	286	Fri	
Flevoland	63	Fri	
Noord-Brabant	417	Fri	
Drenthe	86	Fri	
Zuid-Holland	897	Fri	
Limburg	126	Fri	
Friesland	95	Fri	
Zeeland	75	Fri	
Utrecht	333	Fri	
Groningen	134	Fri	
Overijssel	160	Sat	
Noord-Holland	597	Sat	
Gelderland	265	Sat	
Flevoland	54	Sat	
Drenthe	90	Sat	
Noord-Brabant	350	Sat	
Zuid-Holland	675	Sat	
Limburg	114	Sat	
Friesland	112	Sat	
Zeeland	59	Sat	
Utrecht	287	Sat	
Groningen	109	Sat	

Bar - Victims Age 1 Supplement 1

Victims Age▲	Number casualties✖✖
0 - 4	361
11 - 15	1177
16 - 20	3423
21 - 30	4169
31 - 60	7450
5 - 10	555
61 - 75	2666
76 - Older	1335

Dual axis bar - Day of Accident 1 Supplement 1

Day of Accident ▲	Number casualties ▼	Number accidents ✖✖
Fri	642	549
Thu	636	554
Sat	614	529
Tue	612	518
Wed	572	495
Sun	549	453
Mon	544	454

Hour accident ▲	Number accidents	Number lethal casualties ✖✖
0	43	5
1	99	5
2	75	0
3	59	4
4	58	6
5	46	2
6	80	6
7	131	7
8	192	4
9	169	0
10	111	6
11	117	2
12	149	2
13	171	0
14	177	10
15	225	1
16	252	5
17	305	5
18	269	3
19	201	12
20	171	4
21	129	3
22	132	3
23	127	2
24	64	1

Bar - Victims Age 2 Supplement 1

Victims Age▲	Number casualties✖✖
0 - 4	361
11 - 15	1177
16 - 20	3423
21 - 30	4169
31 - 60	7450
5 - 10	555
61 - 75	2666
76 - Older	1335

Dual axis bar - Day of Accident 2 Supplement 1

Day of Accident ▲	Number casualties ▼	Number accidents ✖✖
Fri	3345	2,965
Thu	3333	3,012
Wed	3145	2,815
Tue	3136	2,856
Mon	2900	2,577
Sat	2872	2,494
Sun	2405	1,974

Hour accident ▲	Number accidents	Number lethal casualties ✖✖
0	134	6
1	259	17
2	205	6
3	153	11
4	151	9
5	132	14
6	232	14
7	586	25
8	1,102	19
9	936	14
10	793	32
11	912	30
12	1,086	32
13	1,166	19
14	1,369	46
15	1,559	44
16	1,565	32
17	1,678	40
18	1,314	24
19	939	30
20	776	24
21	562	13
22	491	11
23	411	15
24	182	5